

WAAR Test Bed for Admitted Students Day

Overview

Test Name: Test Bed or Gyroscope for Admitted Students Day

Subsystem: Hardware

Author: Erika Berreth

Date Executed: 4/3/26

Location: University of Washington

Status: Completed

Credit: The attachment of the drone to the PVC pipe was designed by Diego Heredia.

Objective

The purpose of this testbed is to evaluate the stability and performance of the inner-loop flight controller under control conditions for Admitted Students Day. This will allow students to visually see how the PID controller works to stabilize the drone when manual torque is applied.

Requirements

Requirement	Value	Notes	Status
Degrees of Freedom	≥ 1 DOF (2 DOF target)	Pitch required; roll preferred for coupling effects	MET
Safety Requirement	The system shall include propeller guards or equivalent protective shielding	Prevent user injury and damage during operation	MET (NOTE: we will just not have kids touch the drone).
Disturbance Rejection	The testbed shall allow manual disturbance	Enables evaluation of disturbance rejection	MET

	inputs (e.g., external force applied by hand)		
Mounting	The drone shall be rigidly mounted and capable of withstanding ≥ 1.5 kg vertical thrust force without structural failure	Ensures structural integrity under maximum thrust	MET
Cost	The testbed shall cost less than \$80		MET

Equipment and Setup

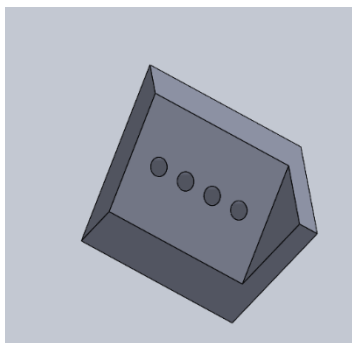
Equipment:

- PLA for the 3D printer (~ 600 grams)
- 5 ft of PVC Pipe
- Wooden plank (~ 4 in x 8 in)
- 8 screws
- Electric Drill + drill bits

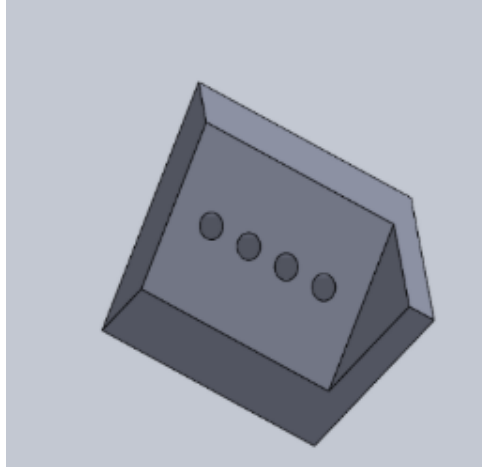
Software:

- SolidWorks

Design Setup:

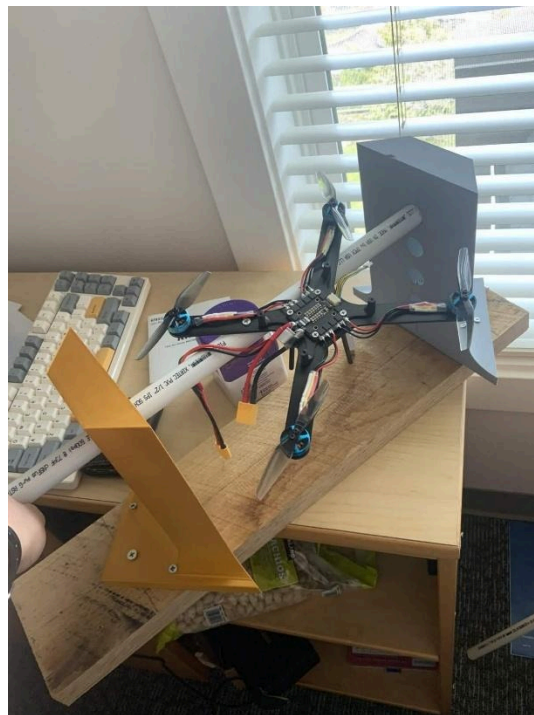


Step 1: Design the side supports in SolidWorks to have a triangular base (for better support) and a trapezoid extension with four holes in the side. See image below for a visual representation.



Tip:

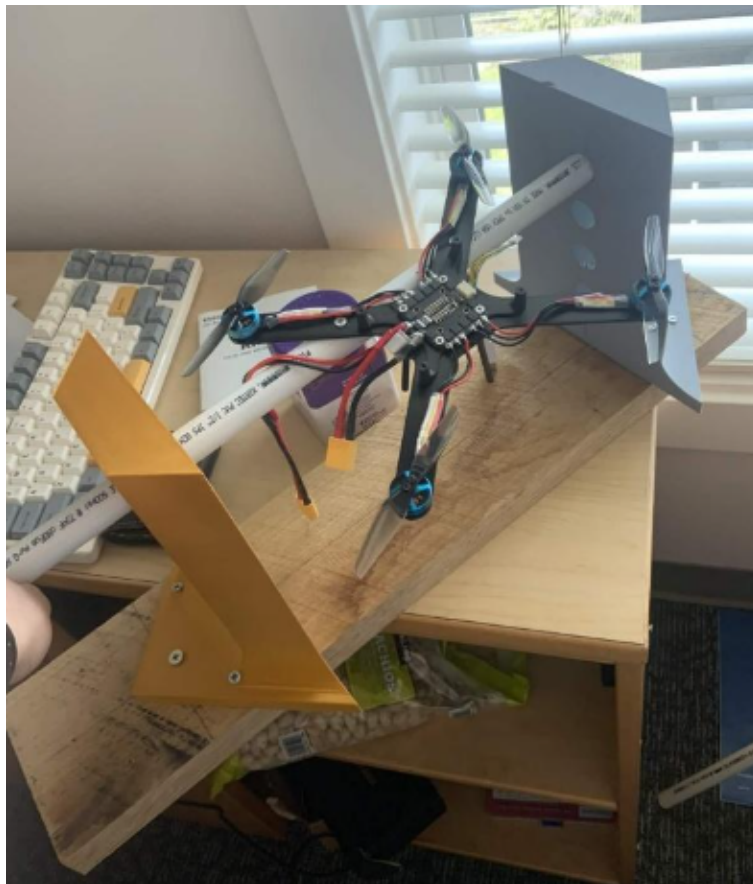
Measure the PVC pipe's cross-sectional dimensions. It was around 0.75 inches for the initial setup with an added tolerance of 0.15. If necessary, cut it to 1.5 inches in length. The height of the structure was around 8.5 inches.



Step 2: Print the side supports with a 3D printer and the PLA.

Step 3: Take the electric drill and with the drill bit drill holes into the plank at three locations that when connected form an isosceles triangle. Then, drill in the screws to attach the side supports to the plank.

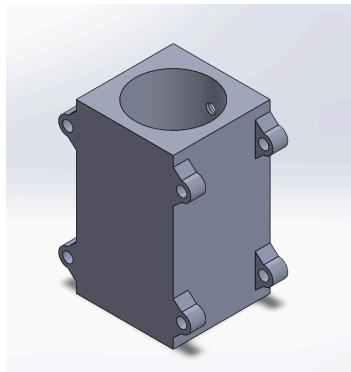
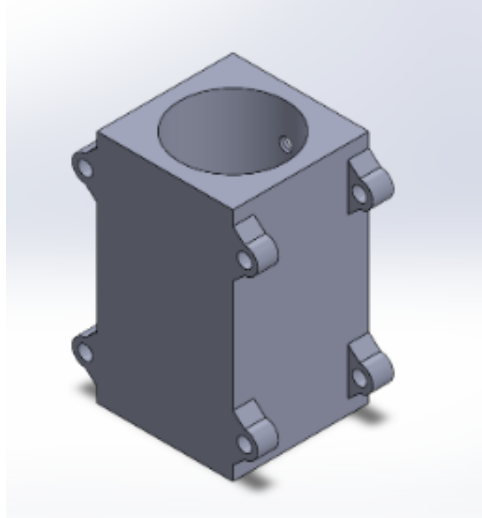
Step 4: Drill a hold on the left and right side of the pipe that goes through the right side support. Then, drill in one screw into each hole. This will prevent the pipe from sliding out of the side supports. An image of the setup with the drone in the frame can be found below.



Tip: Make sure to have more room between the side supports than the longest dimension of the drone.

Step 5: Design the attachment from the drone to the pipe in SolidWorks so that it attaches to the battery mount at the bottom of the drone and has a hole for the PVC pipe to go through.

Tip: The goal of this component is to attach the drone to the pipe so that when someone twists the end of the pipe torque will be applied to the drone. An image of the design can be found below:



Step 6: Print the attachment component with the 3D printer, and use it to attach the drone to the pipe.

Conclusion

Iterations of Design: at first the drone was designed to be connected by a string to the side supports so that it had one degree of freedom. The string would be attached to two ends of the drone frame and to the adjacent side supports so that the drone would be suspended in the air. An image of what this might have looked like can be seen below:



This didn't work for Admitted Students Day because it changed the center of mass of the system. The second iteration of the design involved an attachment to the drone which connected the drone to the pipe so that it was underneath the drone's center of mass and didn't shift it.

Final Test Bed:

